

FACULTÉ DES SCIENCES
Département d'Informatique
Pr. G. BONTEMPI
Campus de la Plaine
Boulevard du Triomphe CP 212
B-1050 Bruxelles
T 02 650 56 14 – F 02 650 56 09
M gianluca.bontempi@ulb.be

To whom it may concern

It is my pleasure to recommend Mr. Olivier Ndinga Oba for admission to a doctoral program. I am a Full Professor at the Université libre de Bruxelles, where I co-head the Machine Learning Group. I taught Mr. Ndinga Oba in my course "Statistical Foundations of Machine Learning" during his Master's degree in Computer Science and Engineering. He was one of the top students in the class of 2025, achieving a final grade of 16.5/20. His work demonstrated a strong understanding of the mathematical foundations of machine learning, including statistical learning theory, model selection, and regularization. He completed his Master's with Distinction.

I have also had the opportunity to review his Master's thesis, entitled "Imitation of Swarm Behaviors with Topological Data Analysis in AutoMoDe-Chocolate and IRACE," which he completed under the supervision of Professor Mauro Birattari at IRIDIA. This work makes a valuable contribution at the intersection of Swarm Intelligence and Topological Data Analysis.

In his thesis, Olivier showed that velocity fields alone are not sufficient for characterizing swarm behaviors, and that adding density fields significantly improves the analysis. He validated this finding through statistical testing using ANOVA and Tukey's HSD. He also developed new metrics based on Wasserstein distances that allow behaviors to be transferred from neural network controllers to finite-state machines. Finally, he extended the AutoMoDe framework by creating a novel schooling behavior module, demonstrating that robotic swarms could successfully achieve complex milling and polarization patterns observed in biological fish schools. This research required a solid grasp of advanced mathematical concepts such as persistent homology, Helmholtz-Hodge decomposition, and multi-scale field analysis, as well as practical programming skills in C++ and Python. I understand that Mr. Ndinga Oba has also recently been a finalist in the Huawei France Tech Arena and in the Future of IT Leaders in Belgium Data & AI Challenge.

Based on his performance in my course and the quality of his research work, I believe Mr. Ndinga Oba has the intellectual ability and technical skills to succeed in a doctoral program. I recommend him without reservation.

Sincerely,



Gianluca Bontempi
Full professor, coHead Machine Learning Group
Computer Science Department
ULB, Université Libre de Bruxelles, Belgium
Web : mlg.ulb.ac.be